

Department of Computer Science and Engineering

Course Number	MATH 243
Course Title	Probability and Statistics
Credit (Contact) Hours	3.0 (3.0)
Level and Term (Section)	Level-2, Term-2
Type of Course	Core Course

Lecture Schedule

Section A Monday (09:00-09:50 am; R- 203) & Wednesday (11.00-11.50 am; R- 203)

Section B Monday (10:00-10:50 am; R- 204) & Wednesday (10.00-10.50 am; R- 204)

Section C Monday (11:00-11:50 am; R- 205) & Wednesday (09.00-09.50 am; R- 205)

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Textbooks Probability and Statistics for Engineers and Scientists by Walpole, Myers, Myers, and Ye, Pearson Education, Inc., Ninth Edition, 2012.

Syllabus

Introduction to Statistics, variability in data, populations and samples, descriptive statistics, inferential statistics and probability, sampling procedures; Measures of location: mean, median; Measures of variability: standard deviation, variance; Higher moments: skewness, kurtosis; Graphical representation of data: scatter plot, stem and leaf plot, histogram, box plot; Probability: sample space and events, rules of probability, conditional probability, independence, Bayes' rule; Random variables: discrete and continuous probability distributions, joint probability distributions, marginal distributions and independence; Expectations, variance and covariance of random variables and their properties, Chebyshev's theorem; Discrete probability distributions: Bernoulli, binomial, multinomial, Poisson distributions and their properties; Continuous probability distributions: uniform, Gaussian (normal), chi-square distributions and their properties; Sampling distributions: sample mean, central limit theorem, sample variance, t-distribution, F-distribution, quantile and probability plots; Statistical inference: parameter estimation, confidence intervals; Hypothesis testing: null and alternative hypotheses, test statistic, P-values and significance levels, Z-test, t-test, goodness-of-fit test; Regression and correlation: least squares, coefficient of determination, correlation coefficient; Analysis of variance (ANOVA).

Lecture#01

Introduction to Probability

Probability: Probability measures uncertainty. Probability gives us a number between **0 and 1** to express how confident we are. 0 means it cannot happen; 1 means it definitely happens. Everything else lies in between. We use it when we cannot predict the exact outcome of an experiment, but we know the range of possible results.

Examples

Toss a coin

We can't say for sure what will happen, but we know the possible outcomes are heads and tails. If the coin is fair, each has probability 0.5.

Rolling a Die

A standard die has six faces. We don't know which number will appear, but the range is fixed.

$$P(\text{rolling a 4}) = 1/6$$

$$P(\text{rolling an even number}) = 3/6 = 0.5$$

Picking a Random Student

In a class of 30 students, 12 are girls and 18 are boys.

$$P(\text{selecting a girl}) = 12/30 = 0.4$$

$$P(\text{selecting a boy}) = 18/30 = 0.6$$

Multiple-Choice Question

A question has 4 options, only one correct.

If a student guesses randomly:

$$P(\text{correct answer}) = 1/4 = 0.25$$

$$P(\text{wrong answer}) = 3/4 = 0.75$$

Related terms to calculate probabilities

Sample Space

The sample space is the set of all possible outcomes of an experiment.

Examples

1. Rolling a single die
 $S = \{1, 2, 3, 4, 5, 6\}$
2. Drawing one card from a standard deck
 $S = \text{all 52 cards}$
3. Taking a newborn baby's sex at birth
 $S = \{\text{Male, Female}\}$
4. Coin: $S = \{H, T\}$

Events

An event is any subset of the sample space.

Examples

1. On a die:
A = rolling an even number
 $A = \{2, 4, 6\}$
2. On a coin:
B = getting heads
 $B = \{H\}$
3. On a card draw:
C = drawing a king
 $C = \{\text{King of Hearts, King of Diamonds, King of Clubs, King of Spades}\}$

Events can contain one outcome or many outcomes.

Set Operations in Probability

Union

$A \cup B$ means A or B happens i.e., $A \cup B$ means elements that are in A or in B or in both.

Example

Rolling a single die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let $A = \{2, 4, 6\}$ (even number)

Let $B = \{1, 2\}$ (number less than 3)

Then

$$A \cup B = \{1, 2, 4, 6\}$$

This is the event of getting an even number or a number less than 3.

Probability rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Intersection

$A \cap B$ means A and B happen together, i.e., $A \cap B$ means elements common to both A and B.

Using the same A and B above:

$$A \cap B = \{2\}$$

Only the outcome 2 is both even and less than 3.

Complement

A^c contains everything in the sample space except A.

Example:

$$A = \{2, 4, 6\}$$

Then

$$A^c = \{1, 3, 5\}$$

This is the event of getting an odd number.

Probability rule:

$$P(A^c) = 1 - P(A).$$

Empty Set $\emptyset$

A set with no elements.

Example

Event $A = \{\text{rolling a 7 on a standard die}\}$.

$$A = \{\emptyset\}$$

Probability of A is zero.

Disjoint Events

Two events are disjoint if they cannot happen at the same time, i.e., two sets that share no elements. In this case, intersection is empty.

Example

On a die:

$$A = \{1\}$$

$$B = \{3\}$$

You can't roll 1 and 3 in the same trial, so A and B are disjoint.

But earlier $A = \{2, 4, 6\}$ and $B = \{1, 2\}$ were not disjoint because they share 2.

Another example

$$A = \{1, 2\}$$

$$B = \{3, 4\}$$

$$A \cap B = \emptyset$$

Probability shortcut for disjoint sets:

$$P(A \cup B) = P(A) + P(B)$$

Probability Rules with Examples

Rule 1: Non-negativity ($P(A) \geq 0$)

$P(A)$ cannot be negative.

Example: $P(\text{rolling a 7 on a standard die}) = 0$. But it's never negative.

Rule 2: Total probability is 1 ($P(S) = 1$)

$$P(S) = 1.$$

Example: For a die,

$$P(1) + P(2) + P(3) + P(4) + P(5) + P(6) = 1.$$

Rule 3: Additivity for disjoint sets

If A and B cannot happen together, i.e., if A and B are disjoint

$$P(A \cup B) = P(A) + P(B).$$

Example

On a die:

$$A = \{1\}$$

$$B = \{2\}$$

$$P(A \cup B) = P(1 \text{ or } 2) = 1/6 + 1/6 = 2/6.$$

General Addition Rule

When events overlap, we subtract the intersection.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example

Let

$$A = \{2, 4, 6\}$$

$$B = \{1, 2\}$$

$$P(A) = 3/6$$

$$P(B) = 2/6$$

$$P(A \cap B) = 1/6$$

So

$$P(A \cup B) = 3/6 + 2/6 - 1/6 = 4/6.$$

Real-Life Example

Suppose you randomly choose a student from a class.

Event A: the student plays cricket.

Event B: the student watches cricket.

Say

$$P(A) = 0.4$$

$$P(B) = 0.7$$

$$P(A \cap B) = 0.35 \text{ (Meaning 35\% both play and watch).}$$

Then

$$P(A \cup B) = 0.4 + 0.7 - 0.35 = 0.75$$

So 75% either play cricket, watch cricket, or do both.